Education & Technical Formation — Revised Knowledge Base

Bachelor of Engineering (BEng) — Healthcare Technology (Sundhedsteknologi)

Technical University of Denmark (DTU), completed 2021

My undergraduate education was built around **full use-case, team-based engineering projects** rather than isolated technical exercises. Across semesters, projects consistently followed an end-to-end model resembling real healthcare technology development:

- Understanding user and clinical needs from a problem description
- Gathering and refining requirements
- Designing solution concepts and user flows
- Validating ideas with users or domain experts
- Iterating based on feedback
- Implementing, testing, and validating solutions

Projects were conducted in teams (typically four people), requiring adaptability rather than strict role separation. While I often took responsibility for the **most technical components** due to team composition (programming, system architecture, data handling), I maintained a working understanding of clinical context, physiology, and user needs, allowing effective collaboration and informed design decisions.

UX and UI design were treated as **integral parts of system design**, not add-ons. We routinely:

- Created paper mock-ups and early concepts for validation before implementation
- Adjusted solution designs after structured feedback
- Used UML diagrams, Gantt charts, milestone planning, and project risk assessments to manage complexity

Through this, I developed an early instinct for **designing systems for people**, even when working on technically demanding components.

Medical Device Development & Regulatory Thinking (Bachelor Level)

A defining course simulated the **full lifecycle of medical device development**, including:

- Direct user needs elicitation (e.g. dialogue with a person living with Type 1 diabetes)
- Risk management and hazard analysis
- Regulatory and compliance considerations
- Understanding how devices are exposed to real-world use, misuse, and edge cases

Rather than separating “cool” and “useful” technology, this course reinforced that **practical constraints, user behavior, and risk exposure fundamentally shape viable designs**. It instilled a lasting awareness that responsibility toward users and patients is inseparable from technical decision-making.

Master of Science in Engineering (MSc) — Biomedical Engineering (Medicin og Teknologi)

Technical University of Denmark (DTU), completed 2023

My master's degree deepened my technical specialization while preserving the same **use-case-driven mindset**. I focused on domains where complexity increases and purely technical mastery becomes insufficient on its own:

- Advanced signal processing in biomedical contexts
- Medical image analysis and machine learning
- Statistical design and analysis of experiments
- Auditory systems, perception, and hearing science

I deliberately pursued machine learning and data analysis electives after my clinical experience at Glostrup Hospital, where I learned how:

- Subtle errors and variability can propagate unnoticed in medical data
- Better analytical tools can uncover hidden structure, uncertainty, and opportunity

My motivation was not algorithmic novelty, but **deeper insight and robustness** in data-driven decision-making.

Audiology as a Deliberate Choice of Complexity

I gravitated toward audiology because it represents a domain where:

- Perception, context, and variability dominate
- The limits of purely technical solutions become explicit
- Biological systems defy clean abstractions

After working with ECG signals and medical imaging—where limitations are often primarily technical—auditory systems forced me to confront **perceptual, physiological, and contextual boundaries**. This reinforced a lasting respect for complexity and for the gap between signal manipulation and human experience.

International Exchange — Hong Kong University of Science and Technology (HKUST)

An international exchange complemented my technical background with perspectives on:

- Human–Computer Interaction
- Regulatory affairs in the healthcare industry
- AI for medical image analysis
- Signal processing and communications

This experience strengthened my appreciation for **usability, regulation, and system context**, and for how cultural and organizational factors influence technology adoption.

Master's Thesis — Auditory Signal Processing (Industry Collaboration)

My master's thesis, conducted in collaboration with an industrial partner, was formative in shaping my professional judgment. During the project, I recognized that the original problem formulation aligned closely with a long-standing, unsolved research challenge (the cocktail-party problem), prompting a necessary reframing of scope.

The experience reinforced that:

- Problem formulation is often the hardest part of engineering
- Biological inspiration does not automatically translate across contexts
- Computational feasibility and perceptual usefulness can diverge sharply

This work trained me to approach complex problems with **critical reflection, methodological humility, and respect for system limits**, rather than forcing solutions.

How This Education Shapes My Professional Approach

Across my education, I consistently learned to:

- Start by understanding **the current situation and its limitations** before proposing solutions
- Treat UX, documentation, and validation as core engineering concerns
- Anticipate sources of variability, error, and risk in data-driven systems
- Balance technical depth with empathy for users and operational reality

This background explains why I naturally operate at the intersection of **engineering, data, healthcare, and user experience**, and why I add the most value in roles that require systems thinking rather than narrow implementation.